Final Project Proposal

Team.

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Elevator Pitch.

- Train a neural network to predict what region a user is from based on the doodling in real time, updating its predictions as drawing continues.

Context.

- Similar to Google's Quick Draw, where the neural network attempts to predict a drawing made by a user
- Want to explore how a user's drawing style can help predict where they are from

Methods. Explain what AI/ML methods you will try to tackle the problem. As opposed to the elevator pitch, here you should be specific about what you will try.

- We will try to train a CNN model to classify a doodle into a region of the world
- The dataset we plan to use, Google Quick Draw, contains country codes, which will be used to compare drawings with the trained model

Data Source.

- Google Quick Draw Dataset: https://github.com/googlecreativelab/quickdraw-dataset
- We will use the NDJSON files, which contain drawing strokes and country-code metadata. We will map country codes into geographic regions and create a manageable subset for training and evaluation.

Code resources. What webpages or generative AIs will you rely on for code? *Please be very frank with us so that we can work to make this a meaningful and interesting project. It is fine to use such resources as long as we're clear on your own contributions (see "What's New" below).*

- *https://github.com/elleryqueenhomels/google_sketcher*
- We will also use the Quick Draw dataset documentation and documentation for our chosen deep-learning framework. We plan to use Claude for help with system design, debugging, unfamiliar concepts, and implementing parts of the drawing interface. We will document any code we adapt or generate and explain our

modifications.

What's new. Explain very clearly what part of the code will come from you instead of the code resources. In addition, mention any additional planned effort on your part that we should consider when deciding whether the project does enough that's new - like gathering a new dataset, for example. *If there isn't enough work mentioned here, your proposal feedback may include additional requirements.*

- We will organize the data into regions, balance the number of drawings from each region, and create images of both finished and unfinished drawings.
- We will build and train the CNN, try different settings, and test how well it predicts regions compared to simpler methods. We will also test whether training with unfinished drawings helps the model make better predictions while someone is still drawing.
- Finally, we will build a drawing interface that updates the model's predictions as the user draws. We will clearly explain which code we wrote ourselves and which parts came from outside sources or AI assistance.

Plan. Describe a reasonable plan to finish your project by the deadline (Dec 8 for Fall '26). ***Include at least one milestone (date & deliverable) that you can present to a TA roughly halfway through the project*.** When planning, keep in mind that you'll also need to do a lightning talk and paper at the end of the semester.

- **October 28–November 6:** Inspect the dataset, select drawing categories and region groups, and create the preprocessing pipeline and data splits.
- **November 7–November 17:** Train an initial CNN and compare it with simple baselines.
- **November 18–November 27:** Generate partial drawings, compare training approaches, and connect the model to a basic drawing interface.
- **December 4–December 8:** Complete the paper, prepare the lightning talk, and finalize the demonstration.

Proposed demonstration or evaluation. How will we know which aspects of your project work well? Describe at least one experiment that would evaluate performance.

- For our demonstration, a user will draw an object on a digital whiteboard. The model will show its guesses about the region and update them as the user continues drawing.
- We will test the model on drawings it has not seen during training. We will measure how often it predicts the correct region and which regions it confuses. We will compare it with a simpler model and with always guessing the most common region.
- We will also test drawings when they are about 25%, 50%, 75%, and 100% complete to see whether more drawing makes the predictions better.

Variations. Identify what you will vary - either to try to achieve the best performance possible, or to see which components of the system are essential.

We will vary:

- **Training images:** Complete drawings only versus a mixture of partial and complete drawings.
- **Model complexity:** A smaller CNN versus a deeper CNN, with dropout to help reduce overfitting.
- **Drawing categories:** Compare performance across individual object categories and when categories are combined.

Have fun brainstorming!